

## Installing the cloud controller

Installation of the cloud controller is very similar to the nodes, with a few exceptions. Once your machine boots from the Eucalyptus Faststart DVD, select the option 'Install CentOS 6 with Eucalyptus Frontend' from the boot screen.

Again, select the appropriate 'Language' and 'Keyboard settings' according to your needs.

Provide a 'Static IP' and a suitable 'Host Name' to your cloud controller in the 'Network Configuration' wizard.

Once done, you will be provided with an interface to supply a 'Public IP Range/ List' (see Figure 4) for your Eucalyptus cloud. You need to enter a valid IP address range here. These public IPs will be mapped to individual Eucalyptus instances (virtual machines) once they are launched in the cloud.

The rest of the installation process remains the same. Once the cloud controller reboots, it will start executing a lot of configuration scripts that will set Walrus, the cluster controller and the storage controller with a few default parameters. The scripts also create a CentOS 6 EMI (Eucalyptus Machine Image). You can use this EMI to launch multiple CentOS 6 instances in your cloud.

To test whether the installation was successful, open a Web browser and type in the following to view the Eucalyptus user console: `http://<Cloud_Controller_IP>:8888`

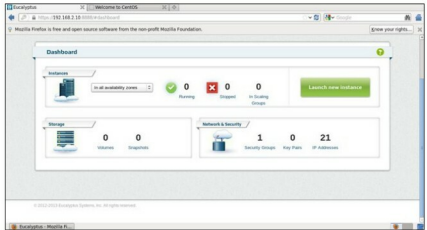


Figure 4: The Eucalyptus user console

- Create a key pair:** A key pair acts as an authentication token between the users and their instances.

To create a key pair, use the following syntax:

```
# euca-create-keypair Key_Pair_Name" >> "Key_Pair_Name".private
```

```
# euca-create-keypair TestKeypair >> TestKeypair.private
```

- Launch the instance:** An instance is launched from a machine image (in this case, EMI). You will require an EMI ID to launch your first instance. This can be obtained from the Eucalyptus cloud user console.

To launch an instance, use the following syntax:

```
# euca-run-instances "EMI_ID" -g "Group_Name" -k "Key_Pair_Name"
```

```
# euca-run-instances "EMI-E90A38DA" -g "TestSecurityGroup" -k "TestKeypair"
```

The instance can take a couple of seconds to launch. You need to note down the instance IP address either from the terminal or by using the Eucalyptus cloud user console. SSH into the instance using the earlier created key pair by using the following syntax:

```
# ssh -i <Key_Pair_Name> ec2-user@<Instance_IP>
```

```
# ssh -i TestKeypair ec2-user@192.168.2.100
```



## Launching your first instance

To launch your first instance, all you need to do is follow these simple steps:

- Create a security group:** A security group is similar to a firewall that protects your instance against network-related attacks. By default, all inbound traffic to the instance is blocked.

To create a security group, use the following syntax:

```
# euca-create-group -d "Description of the group" Group_Name
```

```
# euca-create-group -d "My First Security Group"
TestSecurityGroup
```

Once your group is created, open the necessary ports such as SSH port 22 to obtain access to the instance.

To open a port for your security group, use the following syntax:

```
# euca-authorize -P "Protocol" -p "Port_No" -s "IP_Src" "Group_Name"
```

```
# euca-authorize -P "tcp" -p "22" -s "0.0.0.0/0"
TestSecurityGroup
```

## References

<http://www.eucalyptus.com/eucalyptus-cloud/get-started/try>  
<http://www.eucalyptus.com/docs/eucalyptus/3.3/user-guide/toc.html>

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